

Recitation W5

Problem 1: Community garden planning

(a) Integer programming formulation

Let F be the 10 foods below, $H = \{\text{basil, cilantro, parsley}\}$, and $V = F \setminus H$ (all fruits and vegetables, using the categories in the assignment).

For each food i , let $x_i = 1$ if it is planted and 0 otherwise. Let y_i be the number of plants. Thus $x_i \in \{0, 1\}$ and $y_i \in \mathbb{Z}_{\geq 0}$.

Parameters: s_i is square feet per plant, m_i is the minimum quantity if selected, and e_i is expected harvest in pounds per plant. Following the faculty model, use $M = 100$ in the selection linking constraints.

Food	Category	s_i	m_i	e_i	M
Tomato	Vegetable	4	4	10	100
Bell pepper	Vegetable	2	6	5	100
Carrot	Vegetable	0.5	20	0.25	100
Lettuce	Vegetable	1	10	1.5	100
Strawberry	Fruit	1	12	1	100
Cucumber	Fruit	4	3	8	100
Zucchini	Vegetable	9	2	12	100
Basil	Herb	1	4	1	100
Cilantro	Herb	0.5	6	0.5	100
Parsley	Herb	0.5	4	0.6	100

Objective

Maximize $Z = \sum_{i \in F} e_i y_i$

Equivalently, maximize $10y_T + 5y_B + 0.25y_{Ca} + 1.5y_L + y_S + 8y_{Cu} + 12y_Z + y_{Ba} + 0.5y_{Ci} + 0.6y_P$. Subscripts follow the food names in the table.

The faculty solution is the primary reference for the formulation. Tomatoes are optional. If tomatoes are selected, at least two herb types must also be selected; no separate constraint forces tomatoes to be planted.

(a) Constraints

Space. Total planting area cannot exceed 100 square feet.

$$\sum_{i \in F} s_i y_i \leq 100$$

Selection and minimum quantities. Unselected foods have zero plants. Selected foods meet their minimum quantities.

$$m_i x_i \leq y_i \leq 100 x_i \quad \text{for every } i \in F$$

Variety. Plant at least four different types of food.

$$\sum_{i \in F} x_i \geq 4$$

Herbs require another category. Whenever an herb is selected, at least one fruit or vegetable is selected.

$$3 \sum_{i \in V} x_i - \sum_{h \in H} x_h \geq 0$$

Tomatoes and companion herbs. If tomatoes are selected, include at least two distinct herb types.

$$x_{Ba} + x_{Ci} + x_P \geq 2x_T$$

At least two specified foods. Choose at least two of tomato, bell pepper, and zucchini.

$$x_T + x_B + x_Z \geq 2$$

Zucchini excludes tomatoes. Zucchini and tomatoes cannot both be selected.

$$x_T + x_Z \leq 1$$

Tomato threshold and lettuce. If $y_T > 30$, the first inequality forces $x_L = 1$; the second then requires at least 10 lettuce plants.

$$y_T \leq 30 + 100x_L; \quad y_L \geq 10x_L$$

Variable domains. Plant counts are nonnegative integers; selections are binary.

$$x_i \in \{0, 1\}; \quad y_i \in \mathbb{Z}_{\geq 0} \quad \text{for every } i \in F$$

The tomato threshold is redundant for this garden: $4y_T \leq 100$ implies $y_T \leq 25$. Consequently, more than 30 tomato plants is impossible. The threshold constraint is retained to explicitly represent the stated rule; the lettuce minimum also appears in the general linking constraints.

(b) Excel Solver solution

In the Harvest worksheet, maximize D33 by changing C21:D30. Set C21:C30 to binary and D21:D30 to integer. Use Simplex LP, retain nonnegativity, and set Integer Optimality (%) to 0. Apply the linking constraints and the LHS/RHS checks shown in the screenshots.

Excel Solver returned: “Solver found a solution. All constraints and optimality conditions are satisfied.” An independent integer optimization check gives the same objective value. The table reports the faculty’s selected optimal plan.

Food	Plant? x_i	Plants y_i	Area (sq. ft.)	Harvest (lb.)
Tomato	1	4	16	40
Bell pepper	1	39	78	195
Cilantro	1	6	3	3
Parsley	1	6	3	3.6
Other six foods	0	0	0	0
Total	4	55	100	241.6

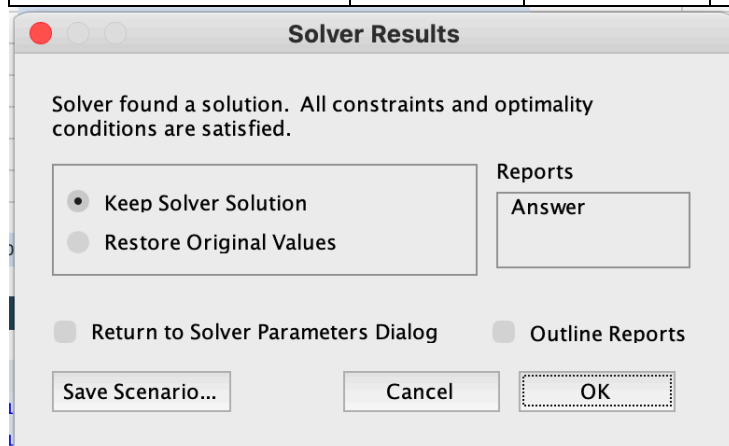


Figure 1. Actual Excel Solver completion dialog for the final model.

(c) Recommendation to the gardening club

Plant 4 tomatoes, 39 bell peppers, 6 cilantro, and 6 parsley. This plan uses all 100 square feet and produces an expected harvest of 241.6 pounds. It includes four food types, supplies the two herbs required by tomatoes, and selects two of the specified foods. Every selected food meets its minimum quantity. Zucchini is excluded, and the lettuce condition is not triggered.

Alternative optima exist. Our rerun also found 20 tomatoes, 7 bell peppers, 6 cilantro, and 6 parsley, yielding the same 241.6 lb. The table and worksheet display the faculty’s plan.

Spreadsheet evidence: parameters and solution

Actual Excel screenshot showing all parameters, both decision-variable columns, the quantity bounds, and the maximized objective value. Space is measured in square feet; expected harvest is in pounds.

	A	B	C	D	E	F	G	H
1		W5 Community garden optimization						
2		PARAMETERS						
3								
4		Food	Category	Space / plant	Minimum	Harvest / plant	Link bound M	
5		Tomato	Vegetable	4	4	10	100	
6		Bell Pepper	Vegetable	2	6	5	100	
7		Carrot	Vegetable	0.5	20	0.25	100	
8		Lettuce	Vegetable	1	10	1.5	100	
9		Strawberry	Fruit	1	12	1	100	
10		Cucumber	Fruit	4	3	8	100	
11		Zucchini	Vegetable	9	2	12	100	
12		Basil	Herb	1	4	1	100	
13		Cilantro	Herb	0.5	6	0.5	100	
14		Parsley	Herb	0.5	4	0.6	100	
15								
16		Space available	100					
17								
18		DECISION VARIABLES						
19								
20		Food	Plant (binary)	Plants (integer)	Min if planted	Max if planted	Space used	Harvest (lb)
21		Tomato	1	4	4	100	16	40
22		Bell Pepper	1	39	6	100	78	195
23		Carrot	0	0	0	0	0	0
24		Lettuce	0	0	0	0	0	0
25		Strawberry	0	0	0	0	0	0
26		Cucumber	0	0	0	0	0	0
27		Zucchini	0	0	0	0	0	0
28		Basil	0	0	0	0	0	0
29		Cilantro	1	6	6	100	3	3
30		Parsley	1	6	4	100	3	3.6
31								
32		OBJECTIVE FUNCTION						
33		Maximize expected harvest (lb)		241.6				

Figure 2. Harvest worksheet, rows 1 through 33.

Key spreadsheet formulas

G5:G14 contains $M = 100$, matching the faculty's linking bound. For the tomato row, $E21 = E5 * C21$ and $F21 = G5 * C21$ give the selected minimum and maximum. $G21 = D5 * D21$ calculates area; $H21 = F5 * D21$ calculates harvest. Copy these formulas through row 30. $D33 = \text{SUM}(H21:H30)$ is the objective.

The template label "Minimize cost" was corrected to "Maximize expected harvest (lb)" to match the assignment.

Spreadsheet evidence: constraints and Solver

	A	B	C	D	E	F	G	H
35	CONSTRAINTS							
36	Constraint	LHS	Relation	RHS	Slack			
37	Planting space		100 ≤		100	0		
38	Different food types		4 ≥		4	0		
39	Herbs need fruit / vegetable		4 ≥		0	4		
40	Tomato companion herbs		0 ≥		0	0		
41	At least two named foods		2 ≥		2	0		
42	Tomato / zucchini exclusion		1 ≤		1	0		
43	Over 30 tomatoes needs lettuce		4 ≤		30	26		
44								
45	Solver: Max D33; change C21:D30; Simplex LP; integer optimality 0%.							
46	Linking: D21:D30 ≥ E21:E30 and D21:D30 ≤ F21:F30.							
47	C21:C30 binary; D21:D30 nonnegative integers; M = 100.							
48	Faculty model: tomatoes optional; at least two herbs if tomatoes selected.							
49								
50								

Figure 3. Every displayed constraint is satisfied. The rowwise quantity bounds appear in Figure 2.

Solver Parameters

Set Objective:

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

-
-
-
-
-
-
-

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Options

Constraint Precision:

☒ Use Automatic Scaling

☐ Show Iteration Results

Solving with Integer Constraints

☐ Ignore Integer Constraints

Integer Optimality (%):

Solving Limits

Max Time (Seconds):

Iterations:

Evolutionary and Integer Constraints:

Max Subproblems:

Max Feasible Solutions:

Figure 4. Actual Solver Parameters and Options dialogs. Integer Optimality is 0%. Tomatoes are optional; the conditional companion-herb requirement is retained.

Primary reference: faculty Recitation W5 Sol.xlsx, Harvest worksheet. Additional sources: RS W5.pdf, Problem 1, and Recitation W5.xlsx.